Computer Security — Quick Notes

Short answers to the questions from our discussion

1. Hashing vs. digital signatures

Hashing turns data into a fixed-length fingerprint. Comparing hashes helps detect changes, as long as the expected hash itself is trustworthy. A **digital signature** also helps detect changes, but additionally provides evidence that the data was signed using a particular private key. Signatures typically use hashing internally.

2. Can two different messages have the same hash?

Yes. This is a **collision**. Because there are far more possible inputs than fixed-size hash outputs, collisions must exist. Secure cryptographic hashes such as SHA-256 are designed to make finding one infeasibly difficult in practice.

3. Private key and public key in signatures

The **private key signs**; the matching **public key verifies**. A valid signature supports the claim that the data was signed by whoever controls that private key and that the signed data has not changed. It does not prove who physically sent the message, and it does not hide the contents.

4. Digital certificates

A certificate links an identity (such as a domain name) to a public key and is signed by a certificate authority (CA). Browsers validate the certificate chain against trusted root certificates and check details such as domain name and expiration. A valid certificate does not guarantee that a website is honest or safe.

5. Botnets and owned devices

A **botnet** is a coordinated group of devices, commonly compromised and remotely controlled, used for tasks such as DDoS attacks. Someone could also use hundreds of their own Raspberry Pis to flood a service; that is a coordinated attack, though the devices are not necessarily called a botnet if they are not compromised or operating as bots in the usual sense.

6. Authentication, access control, and confidentiality

Authentication — "Who are you?" Checks identity, e.g. logging into a university portal.

Access control — "What are you allowed to do?" Grants or denies actions, e.g. a student can view but cannot edit grades.

Confidentiality — "Who should be able to see this information?" The goal of preventing unauthorized disclosure. Access control helps achieve it; encryption can also protect confidentiality.

Remember: Authentication identifies you; access control limits your actions; confidentiality protects information from unauthorized disclosure.